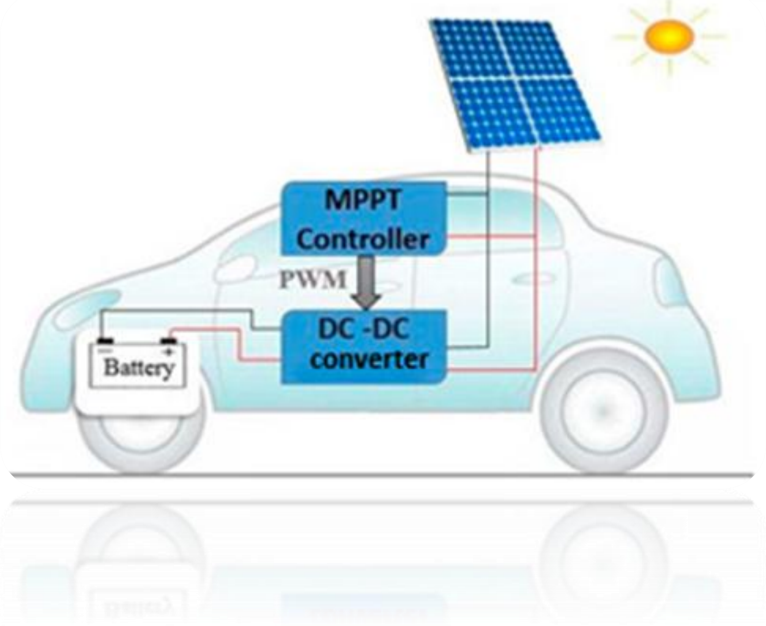
PV Integrated EV System

Next-Generation 48V $LiFePO_4$ & Supercapacitor Infrastructure for EV Charging

System Architecture & Principle

- **Solar Input:** Photovoltaic panels convert sunlight to DC electricity (18–36V open circuit).
- **MPPT Controller:** Maximum Power Point Tracking matches the 48V $LiFePO_4$ charging profile.
- **Supercap Buffer:** Module absorbs rapid current surges from cloud-shade cycling.
- **Battery Pack:** 48V system (15S configuration) with integrated BMS protection.



PV Panel Selection: Mono-Crystalline

Higher Efficiency

Superior conversion of irradiance to electrical output under standard and low-light conditions.

Lightweight & Flexible

Conforms to curved or irregular rooftop surfaces without heavy mounting hardware.

Lower Degradation

Longer operational lifespan with minimal performance loss over time under environmental stress.

Estimated Rooftop Output Power

Parameter	Value
Solar Irradiance (G)	1 kW/m^2
Panel Efficiency (η)	20%
Estimated Area (A)	1.5 m^2

Power Calculation

$$P = A \times G \times \eta$$

$$P = 1.5 \times 1000 \times 0.20 = 300W$$

Maximum Power Point Tracking

Sense: Measures PV voltage and current continuously to monitor real-time output.

Track: Advanced algorithms (e.g., P&O) find the maximum power point on the P-V curve.

Adjust: Changes boost converter duty cycle to maintain the MPPT despite solar variation.

Ensures system operates at peak efficiency during changing irradiance, temperature, and partial shading.

DC-DC Boost Converter Hardware

Specifications

- High-Speed MOSFET Swich
- Fast Recovery Power Diode
- Switching Frequency: 20-100 kHz

System Assumptions

- Vin (PV): 18V – 36V
- Vout (Battery): 48V – 58.4V
- Estimated Efficiency: 90% – 95%

Supercapacitor Buffer Strategy

Surge Protection & Lifecycle

An EDLC (Electric Double-Layer Capacitor) module placed between MPPT and Battery:

- Reduces battery stress by 20–30%
- Handles rapid cloud-shade cycling
- Prevents $LiFePO_4$ micro-charging cycles

DC-DC boost converter calculation

Equation used

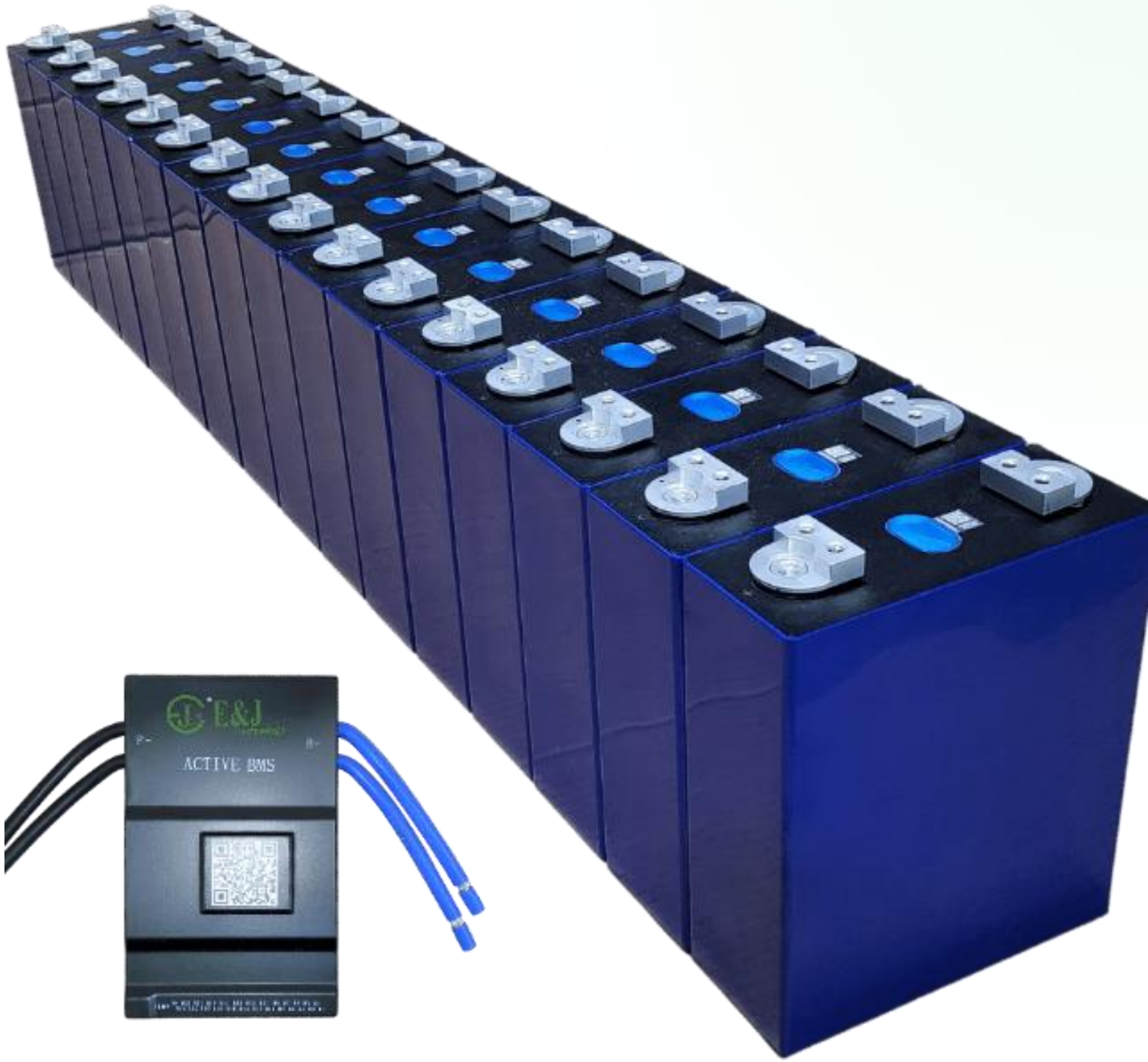
$$D_{max} = 1 - \frac{V_{in(min)}}{V_{out}}$$

$$L_{min} = \frac{V_{in(min)} \cdot D_{max}}{\Delta I_L \cdot f_{sw}}$$

$$C_{out} = \frac{I_{out(max)} \cdot D_{max}}{f_{sw} \cdot \Delta V_{out}}$$

48V $LiFePO_4$ Solution

- **Safe & Stable:** No thermal runaway risk for lab or field use.
- **Solar-Ready:** 48V standard for MPPT controllers.
- **Long Life:** 2000–4000+ cycles for years of daily use.
- **Budget-Friendly:** 15S pack utilizes widely available 3.2V cells.



EV Battery Voltage Benchmarks

Vehicle Type	Voltage	Capacity	Chemistry
E-bicycle	24–48 V	10–20 Ah	$LiFePO_4$
E-scooter	48–72 V	24–60 Ah	Li-ion/NMC
E-rickshaw (3-whl)	48–60 V	100–150 Ah	LFP / Lead-acid
Mid-range EV car	350–400 V	40–82 kWh	NMC 811
Premium 800V EV	800 V	77–100 kWh	NMC 811
Electric City Bus	500–700 V	200–400 kWh	LFP

Energy Generation & Driving Range

1.5

kWh / Day Yield

18.7

Daily Km Range Added

The Efficiency Context

Based on 5 peak sun hours in Sri Lanka and a three-wheeler EV consuming **80 Wh/km**. The solar integration provides a significant daily offset for urban commutes.

Prototype Selection & Cost

Configuration	Cells	Spec	Cost (LKR)
15S1P Basic Demo	15 Cells	48V / 6Ah (288 Wh)	**19,350**
Cell Type	32700	3.2V 6000mAh (5C)	1,290 / cell
Peripheral Comp.	BMS, MPPT	Solar-Ready Module	*TBC*

*Peripheral component pricing for BMS, MPPT and Supercapacitor modules to be finalised on purchasing of batch.